Intelligent Optimization

Tabu Search Module Homework 5, Due March 20, midnight

For the 15 department QAP you already studied.

<u>Tabu Search</u>

1. Code a simple TS to solve the problem. To do this you need to encode the problem as a permutation, define a neighborhood and a move operator, define the tabu list entry, set a tabu list size and select a stopping criterion. Use only a recency based tabu list and no aspiration criteria at this point.

2. Run your TS using 5 different random starting points.

3. Perform the following changes on your TS code and compare the results.
   - change the tabu list size - smaller and larger than your original choice (one smaller size and one larger size). Do this with the same 5 random starting points as above (10 more runs).
   - change the tabu list size to a dynamic one - an easy way to do this is to choose a range and generate a random uniform integer between this range every so often (i.e., only change the tabu list size infrequently). Do this once for the same 5 random starting points (5 more runs).
   - add one or more aspiration criteria such as best solution so far, or best solution in the neighborhood, or in a number of iterations and use the same 5 random starting points and one of the tabu list sizes from above (or the dynamic tabu list size). (5 more runs).
   - use less than the whole neighborhood (such as best of a randomly chosen 50%) or first improving) to select the next solution and try this for TS from the immediate above bullet (aspiration and a chosen tabu list size) for the same 5 starting points (5 more runs).
   - add a diversification mechanism such as either a frequency based tabu list and/or aspiration criteria (designed to encourage the search to diversify) or a restart mechanism, and use with the TS from two bullets above (aspiration and a chosen tabu list size for the whole neighborhood) for the same 5 starting points (5 more runs).

This will be a total of 35 runs.

4. Please include your code with your homework.

This is the sixth of the QAP (quadratic assignment problem) test problems of Nugent et al.* Fifteen departments are to be placed in 15 locations with five departments (columns) in three rows of five departments in each row. The objective is to minimize flow costs between the placed departments. The flow cost is (flow * distance), where both flow and distance are symmetric between any given pair of departments. Below is the flow (lower half) between departments and rectilinear distance (upper half) between locations matrix. The optimal solution is 575 (or 1150 if you double the flows).

* C. E. Nugent, T. E. Vollmann and J. Ruml, An experimental comparison of techniques for the assignment of facilities to locations. *Operations Research* 16, 150-173 (1968)

Dept	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	-	1	2	3	4	1	2	3	4	5	2	3	4	5	6
2	10	-	1	2	3	2	1	2	3	4	3	2	3	4	5
3	0	1	-	1	2	3	2	1	2	3	4	3	2	3	4
4	5	3	10	-	1	4	3	2	1	2	5	4	3	2	3
5	1	2	2	1	-	5	4	3	2	1	6	5	4	3	2
6	0	2	0	1	3	-	1	2	3	4	1	2	3	4	5
7	1	2	2	5	5	2	-	1	2	3	2	1	2	3	4
8	2	3	5	0	5	2	6	-	1	2	3	2	1	2	3
9	2	2	4	0	5	1	0	5	-	1	4	3	2	1	2
10	2	0	5	2	1	5	1	2	0	-	5	4	3	2	1
11	2	2	2	1	0	0	5	10	10	0	-	1	2	3	4
12	0	0	2	0	3	0	5	0	5	4	5	-	1	2	3
13	4	10	5	2	0	2	5	5	10	0	0	3	-	1	2
14	0	5	5	5	5	5	1	0	0	0	5	3	10	-	1
15	0	0	5	0	5	10	0	0	2	5	0	0	2	4	-

<u>Extra Credit</u>: for a possible extra 25 points, repeat all of these but change the entry to the tabu list to something quite different. Comment on the effect of the tabu search entry.